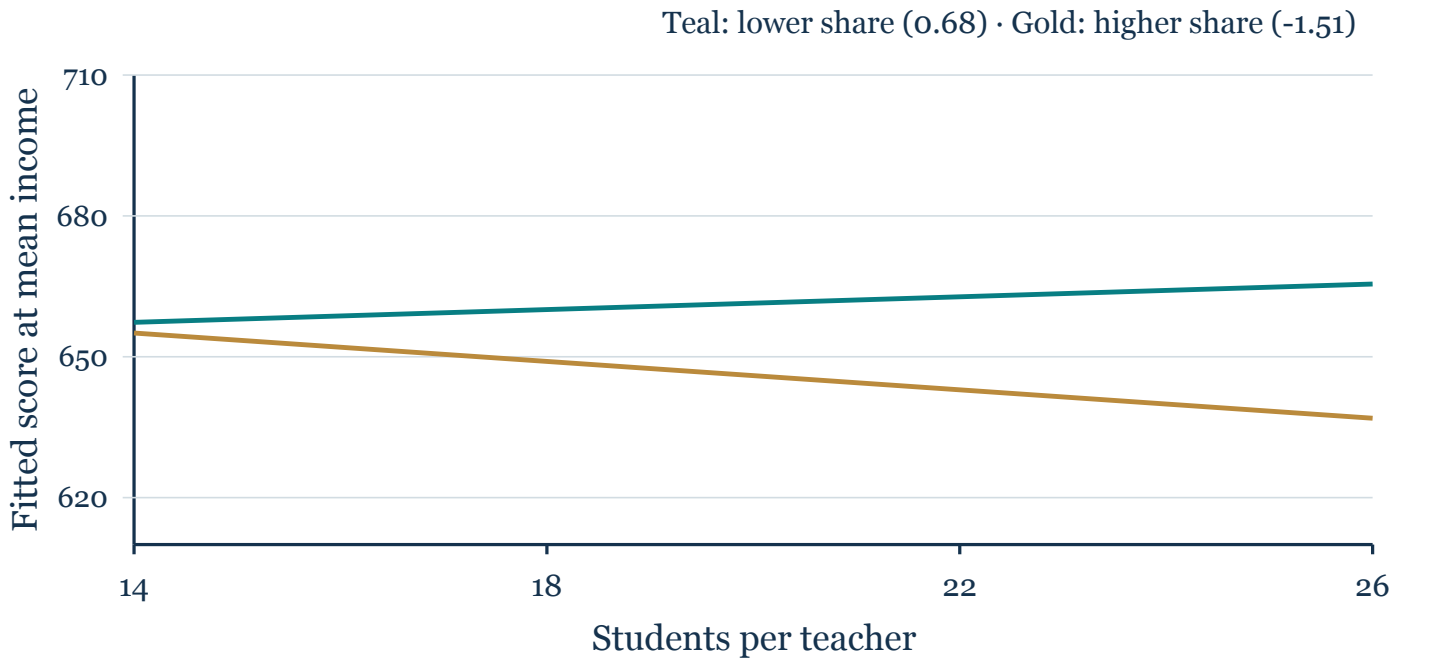
Does the staffing association differ across district groups?

An interaction lets slopes differ in a stated way.



Actual group-specific fitted lines at the overall sample mean of income. High language share means at least 10 percent English learners.

Define the indicator and reference group

Coding determines which coefficients describe the baseline.

> $D_i = 1\{\text{EL}_i \geq 10\}$

Write the interacted equation

Include the main terms as well as their product.

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 X_i D_i + \gamma W_i + u_i$$

γ: common income slope; XD: staffing–group product

Substitute zero and one before interpreting

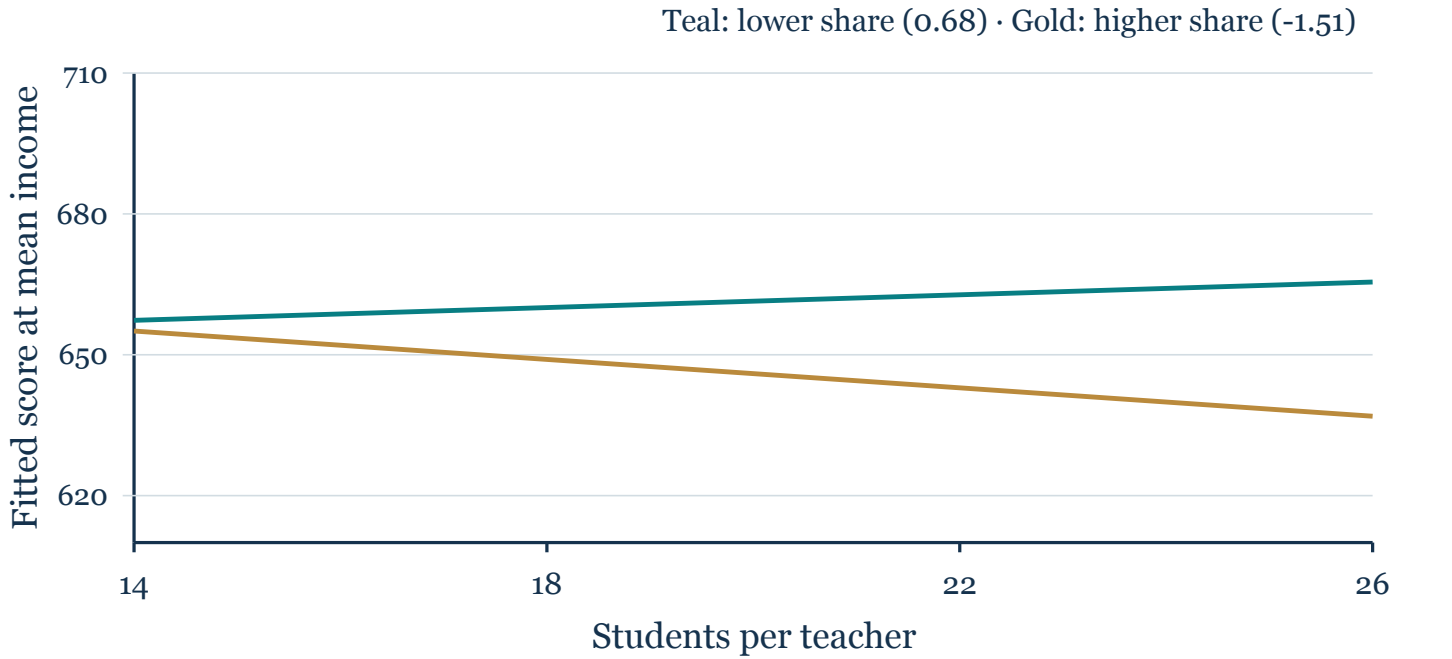
Two group equations reveal the relevant combinations.

> $D = 0: \quad \partial E(Y \mid X, D, W)/\partial X = \beta_1$

> $D = 1: \quad \partial E(Y \mid X, D, W)/\partial X = \beta_1 + \beta_3$

Read both fitted lines in the actual school data

The reference slope and slope difference can have different signs.



Actual group-specific fitted lines at the overall sample mean of income. High language share means at least 10 percent English learners.

Interaction contrast

Apply the idea before reading the solution.

Practice

In $Y = a + 0.4X - 0.9(X \times near) + c\,near + u$, near is 0 or 1. What are the x slopes far from and near the border?

Interaction contrast: explanation

Connect the calculation to the empirical claim.

> Far: 0.4. Near: $0.4 - 0.9 = -0.5$. The interaction is the difference in slopes, −0.9. The near-group intercept shift c is a different quantity.

Factor-variable syntax keeps the model explicit

Use Stata's interaction notation to include the needed terms.

```
. regress score c.str##i.high income, vce(robust)

------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |   .6800597   .3532615     1.93   0.055    -.0143453    1.374465
      1.high |   28.41596   12.83889     2.21   0.027     3.178593    53.65332
             |
  high#c.str |
         1   |  -2.192258   .6399264    -3.43   0.001    -3.450159   -.9343566
             |
      income |   1.651672   .1034089    15.97   0.000     1.448401    1.854942
       _cons |   622.5112   7.209389    86.35   0.000     608.3397    636.6826
------------------------------------------------------------------------------
```

Calculate a policy contrast within a group

A two-student change uses that group's total slope.

```
. lincom -2*str-2*1.high#c.str

------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   3.024396   1.071979     2.82   0.005       .91721    5.131582
------------------------------------------------------------------------------
```

$$\Delta\widehat{Y}_{D=1} = -2(\hat{\beta}_1 + \hat{\beta}_3)$$

The SE of a sum needs covariance

Coefficient estimates can move together across samples.

> $\mathrm{Var}(\widehat{\beta}_1 + \widehat{\beta}_3) = V_{11} + V_{33} + 2V_{13}$

> V_{11}, V_{33}: variances; V_{13}: covariance

Read the higher-group slope as one tested quantity

Use the output for the combination the question asks about.

```
. lincom str+1.high#c.str

------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |  -1.512198   .5359897    -2.82   0.005    -2.565791    -.458605
------------------------------------------------------------------------------
```

A group gap can depend on X

The group main effect is not the gap at every staffing ratio.

> $\text{Gap}(X) = \beta_2 + \beta_3 X$

A joint hypothesis asks whether several restrictions hold together

One coefficient test cannot replace the combined question.

```
. test 1.high 1.high#c.str

 ( 1)  1.high = 0
 ( 2)  1.high#c.str = 0

       F(  2,   415) =   98.88
            Prob > F =    0.0000
```

> $H_0: \ \beta_2 = 0, \ \beta_3 = 0$

Uncertainty of a sum

Apply the idea before reading the solution.

Practice

To test the near-group slope $\beta_1 + \beta_2$, is $\mathrm{SE}(\widehat{\beta}_1) + \mathrm{SE}(\widehat{\beta}_2)$ correct?

Uncertainty of a sum: explanation

Connect the calculation to the empirical claim.

> No. Its variance is $\mathrm{Var}(\widehat{\beta}_1) + \mathrm{Var}(\widehat{\beta}_2) + 2\,\mathrm{Cov}(\widehat{\beta}_1, \widehat{\beta}_2)$. Take the square root of that sum; use lincom after the fitted interaction model. The covariance cannot generally be ignored.

Classical fit-loss F tests require matched models

Restricted and unrestricted fits must answer one nested comparison.

> $F = \dfrac{(SSR_R - SSR_U)/q}{SSR_U/(n - k_U)}$ under classical conditions

> SSR: residual sum of squares; q: restrictions; k_U: unrestricted coefficients

Centering changes coefficient labels, not fitted values

Choose a reference value that helps interpretation.

> $X_i^c = X_i - 20$

Check whether fixed effects absorb a regressor

A time-only variable cannot survive unrestricted time indicators.

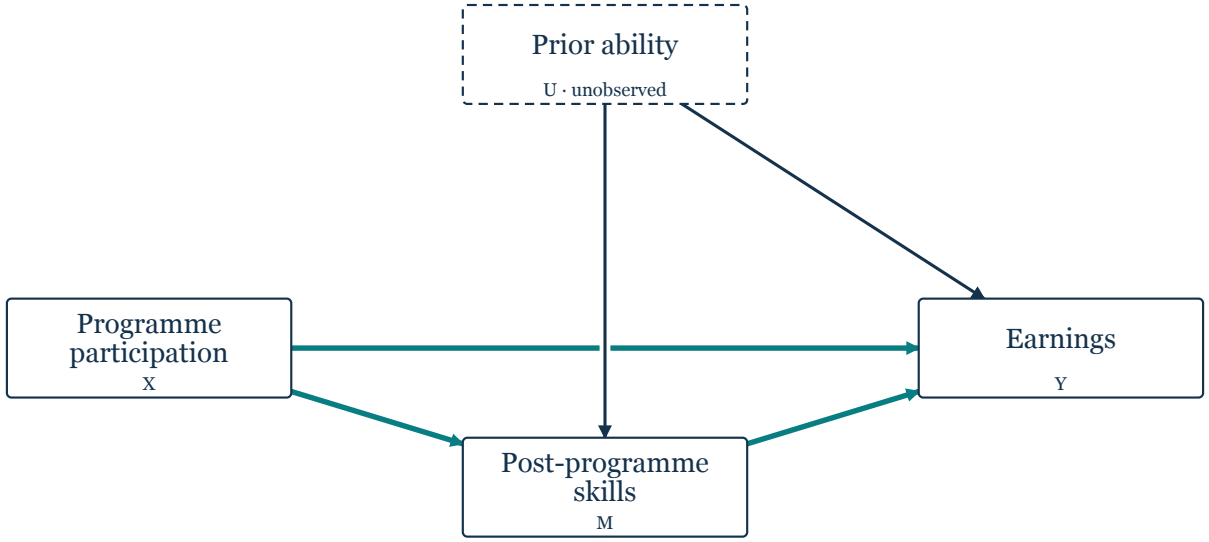
> $T_t \in \text{span}\{\text{year indicators}\}$

> $T_t D_i$ can retain two-way residual variation

> span: all linear combinations of the listed variables

Separate group heterogeneity from causal identification

A different slope is still a fitted relationship.



2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Choose the dependence unit for inference

The number of rows need not equal the number of independent sources.

> If several observations share a policy assignment or common shocks, their errors may be related.

> For a state policy observed in many districts, state-level clustering may be a starting point if states provide the relevant independent variation.

Build an empirical comparison before reading stars

List the coefficient combination and its units.

> For each question, write the required expression: a reference slope, a second-group slope, a slope difference or a group gap at a stated X.

> In a policy memo, report a feasible change within each group and the evidence for any difference between them.

What year effects absorb

Apply the idea before reading the solution.

Practice

Tax_t is identical across all districts in each year. Can its coefficient be estimated separately in a regression with an intercept and a full set of year effects? Can $Tax_t \times near_i$ survive?

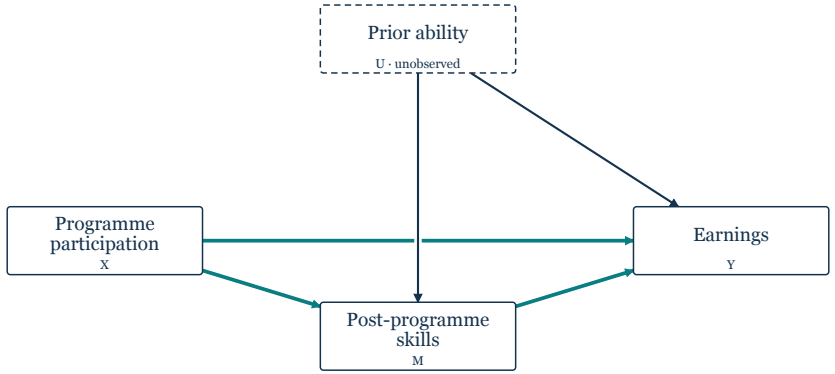
What year effects absorb: explanation

Connect the calculation to the empirical claim.

> The common tax variable is an exact linear combination of year indicators and is absorbed. The interaction can survive because it differs across groups within a year and changes over time. With both unit and year effects, identification requires nonzero two-way residualized interaction variation. It estimates a differential slope, not the absorbed common effect.

DAG practice: a binary treatment still needs a control argument

Trace the routes before choosing a control.



2 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Programme participation affects post-programme skills and earnings. Prior ability affects skills and earnings. If the programme is initially independent of ability, what two changes can occur when skills are held fixed? Would adding a group interaction automatically resolve them?

DAG practice: a binary treatment still needs a control argument: explanation

Explain what conditioning changes in this stated graph.

> Conditioning on skills blocks the mediated programme route $X \to M \to Y$. It can also open $X \to M \leftarrow U \to Y$, where skills are a collider and ability influences earnings. A regression can consequently mix a changed target with new selection. A group interaction changes the specified shape or subgroup slope. It does not automatically supply the missing causal control argument. Define the intervention and whether its total or a suitable direct effect is wanted, then assess assignment and mediator–outcome assumptions before interpreting the interaction causally.

Turn an interaction into two policy contrasts

Reproduce the evidence, then explain its economic meaning.

Practice

Run meeting_07.do. Compute the staffing slope for each language-share group and the fitted score contrast for a reduction of two students per teacher in each group, holding income fixed. Use the lincom output to assess the higher-share contrast. Explain why the group slope difference and the group intercept difference answer different questions.

Turn an interaction into two policy contrasts: explanation

A reproducible result needs both a calculation and an interpretation.

> The lower-share staffing slope is 0.680060. Adding the interaction coefficient -2.192258 gives the higher-share slope -1.512198. The fitted two-student contrasts are therefore -1.3601 and 3.0244 score points, respectively. Use lincom for the second contrast because its uncertainty depends on covariance between the staffing and interaction estimates. The interaction measures a difference in slopes, while the group indicator shifts the intercept. Neither is automatically a causal subgroup treatment effect. The adjusted comparison and the reason for different staffing choices within each group still need to be defended.

Carry the same reasoning into ordered observations

Time adds structure to both models and uncertainty.

> We can now represent different group slopes, calculate combined effects and evaluate joint restrictions.

> The next teaching block turns to observations ordered through time.